

Welcome to STAT 8025

Lec 1: Introduction and Overview

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Instructor

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- ▶ Relevant course material will be posted on Canvas

- ▶ Student Information:
 - ▶ Name: Last, First
 - ▶ Major
 - ▶ Graduate-level statistical courses taken
 - ▶ Signature and Date
- ▶ **Homework 0:** Read the syllabus carefully, fill in the blanks and return the last page by January 15 Friday!

Introduction

- ▶ First courses in Statistics...
 - ▶ “Assume the data are a **simple random sample** from the population”
 - ▶ “Let $X_1, X_2, \dots, X_n \stackrel{iid}{\sim} F_\theta$ ”
- ▶ Most data are NOT independent; they are **dependent**
Types of dependent data?

Introduction

“After choosing the *area* we usually have no guidance beyond the widely verifiable fact that patches in **close proximity** are commonly **more alike**, as judged by the yield of crops, than those which are far apart.”

– R. A. Fisher (1935) on analyzing agricultural field

First Law of Geography

“Everything is related to everything else, but **near** things are **more related** than distant things.”

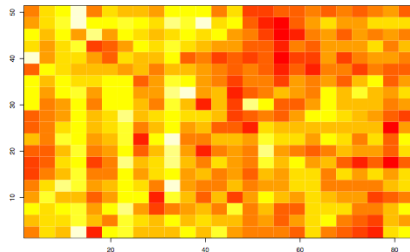
– W. Tobler (1970) *Economic Geography*

Why Spatial Statistics, ctd

- ▶ Classical **independent** statistical assumption
 - ▶ The values of observations in each sample are usually assumed to be independent of one another
 - ▶ Correlated samples contain _____ (less/equal/more) information than the same number of independent samples
- ▶ **Spatial Statistics** broadens the classical statistical models and methods to those **recognize the presence and importance of spatial information**
 - ▶ The 'location' of data need to be considered in statistical analyses.
 - ▶ *Spatial dependence* needs to be accounted for or exploited.
- ▶ A vast subject due to complexity of spatial data
 - ▶ Observations may be *univariate or multivariate, categorical or continuous, etc.*
 - ▶ Locations may be *points or regions, regularly or irregularly spaced or shaped (and even random)*

A BRIEF HISTORY

- ▶ Experiment design and agriculture field trials
- ▶ Spatial variation in wheat yield could be due to variation in the microenvironment.



(Mercer and Hall Wheat Yield Data, 1911)

- ▶ Taking a statistical approach, we could assume this variation is stochastic and write down the following model:

$$Y_{ij} = \mu + Z_{ij},$$

for $i = 1, \dots, 20$, $j = 1, \dots, 25$ (block); Z_{ij} are mutually independent, zero-mean random variables

- ▶ Why is this not a good model for the Mercer and Hall data?



“.. the widely verifiable fact that patches in **close proximity** are commonly **more alike**, as judged by the yield of crops, than those which are far apart.”

– R. A. Fisher (1935)

► Add a blocking factor:

$$Y_{ij} = \mu + \beta_j + Z_{ij},$$

where $\sum \beta_j = 0$.

► A **model-based** solution

“.. Now in general the correlation weakens as the unit of time or space grows larger and I can't help thinking that it would be a great thing to work out the law according to which the correlation is likely to weaken with increase of unit..”

– Wm. Gosset to K. Pearson (1910)

- ▶ Modeling spatial variation in geological and environmental processes
 - ▶ Mining engineering
 - ▶ Witwatersrand, South Africa - gold mining (Krige)
 - ▶ Centre de Geostatistique, L' École des Mines, Fontainebleau, France - geostatistics (Matheron and others)



► Forestry

► Matérn family of covariance functions

$$\rho(u) = \text{cor}(X(s), X(s')) = [2^{\kappa-1} \Gamma(\kappa)]^{-1} (u/\phi)^{\kappa} K_{\kappa}(u/\phi)$$

► Emergence of modern spatial statistics

1. Besag (1974) – “Spatial interaction and the statistical analysis of lattice systems”

Contribution: space is different than time; self-consistent full conditional distributions

2. Ripley (1977) – “Modelling spatial patterns” Contribution: compare simulated realizations from a proposed model to functional summary statistics of the data

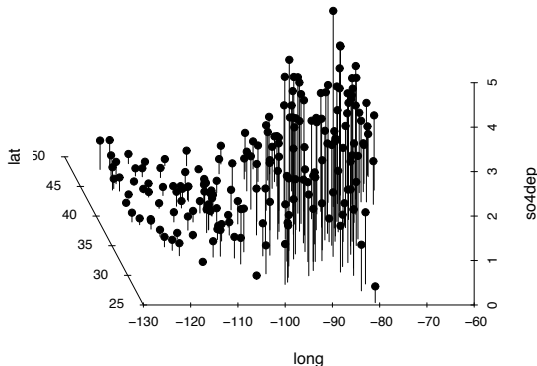
3. Cressie (1993): *Statistics for spatial data*

Contribution: unified, rigorous treatment of the subdiscipline

► State-of-the-art research and practice: hierarchically-specified models, massive data, fast computation, space-time, multivariate, non-Gaussian, etc.

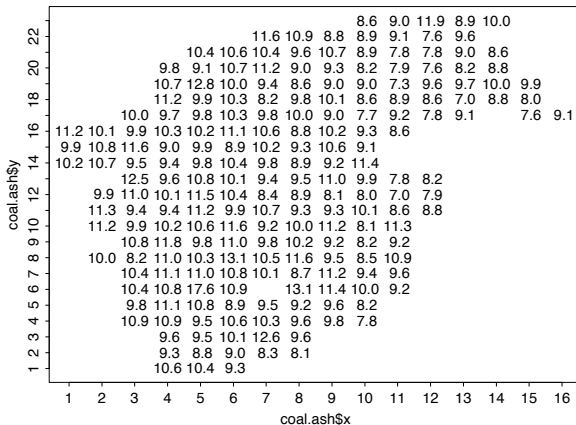
Spatial Datasets

Wet deposition of SO_4 (g/m^2) in 1987 at National Acid Deposition Program sites.



Spatial Datasets

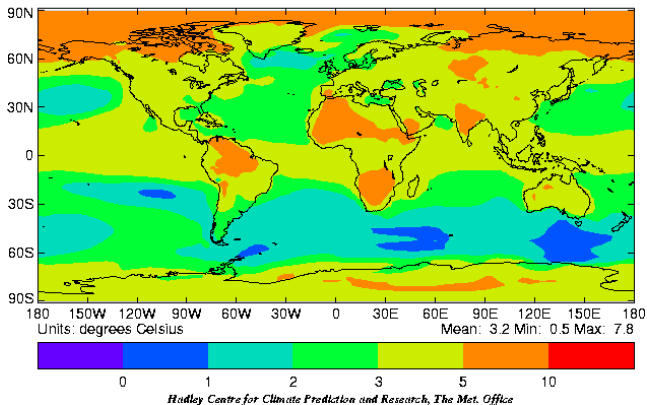
Coal ash samples from a mine in Pennsylvania



Spatial Datasets

Climate change (Source: Hadley Center)

Change in annual average surface air temperature
from 1960–1990 to 2070–2100 from HadCM2 IS92a



Real estate (Source: www.Zillow.com)

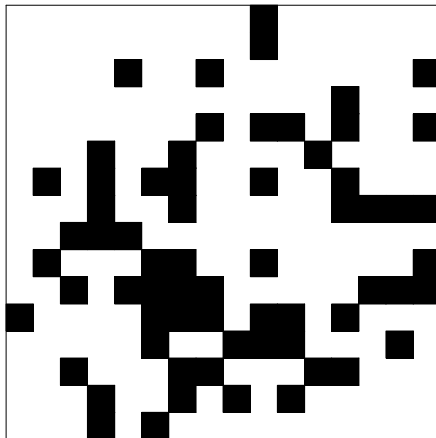


Periodontal disease



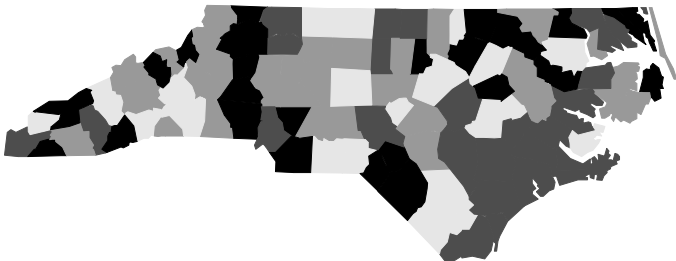
Spatial Datasets

Presence (black) or absence (white) of *Atriplex hymenelytra* on a grid of quadrats in Death Valley, CA.



Spatial Datasets

Population-adjusted mortality rates due to SIDS in counties of North Carolina, 1974- 1978.



Components of Spatial Data

- ▶ **Features:** point, line, area, volume at specified locations
- ▶ **Attributes:** observations associated with features
- ▶ In Spatial Statistics, we study not only the attributes of spatial data but also the features with these data
- ▶ In Spatial Statistics, maps are often used to present the data and the analysis results

Issues with Spatial Data

Describing Location and Distances On the globe

- ▶ Latitude (ϕ) and Longitude (λ)
- ▶ The earth is round! so (longitude, latitude) $\neq$ (x, y) when we compute distances!
- ▶ **Great circle distance**: Consider two points on the surface of the earth, $s_1 = (\lambda_1, \phi_1)$ and $s_2 = (\lambda_2, \phi_2)$. The shortest distance (geodesic) between them on the surface of the earth is called the great circle distance, which is

$$D = R \arccos[\cos\phi_1 \cos\phi_2 \cos(\lambda_1 - \lambda_2) + \sin(\phi_1) \sin(\phi_2)]$$

where R is the earth radius

Fundamental of Cartography: Map Projections

- ▶ A **map projection** is a systematic representation of all or part of the surface of the earth on a plane
- ▶ Theorem: The sphere cannot be flattened onto a plane without distortion
- ▶ Instead, people use an intermediate surface that can be flattened. The sphere is first projected onto the this **developable surface**, which is then laid out as a plane
- ▶ The three most commonly used surfaces are the **cylinder**, the **cone**, and the **plane** itself. Using different orientations of these surfaces lead to different classes of map projections...

What is Spatial Statistics?

Several descriptions can be found in the literature

- ▶ The combination of **attribute information** with **spatial information** and their **statistical analysis**
- ▶ Use spatial information to help model the variability of attributes
- ▶ Modifications, extensions, and additions to statistical techniques that explicitly *consider the importance of locations or spatial arrangement*

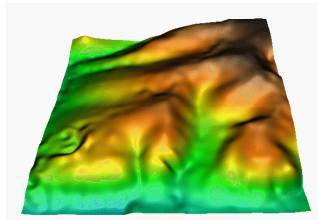
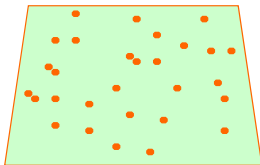
Basic Ingredients of Spatial Statistics

- ▶ **Observations** on one or more “response” variables (attributes) are taken at multiple, identifiable sites in some spatial domain
- ▶ **Locations** of these sites are observed and are attached, as labels, to the observations
- ▶ An analysis of the observations is performed, in which the **spatial locations of sites are taken into account**
- ▶ Either the observations or the spatial locations (or both) are modelled as random variables, and **inferences** are made about these models and/or **about additional unobserved/unknown variables/parameters**

Types of Spatial Data

Geostatistical Data

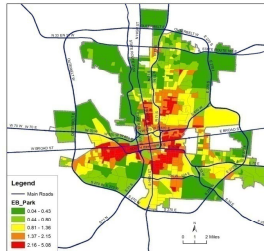
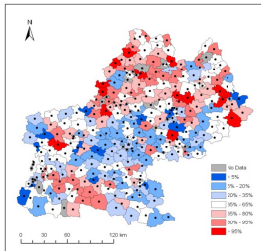
- ▶ **Geostatistics**: an area of spatial statistics concentrating on spatial mapping of attributes in **fixed continuous (uncountable)** space D
- ▶ The attributes are the random variables of interests
- ▶ The attributes exist at infinitely many locations but only observed at a finite number of locations



Types of Spatial Data

Lattice Data

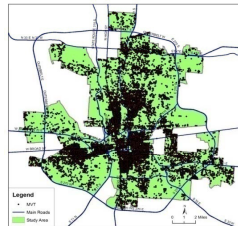
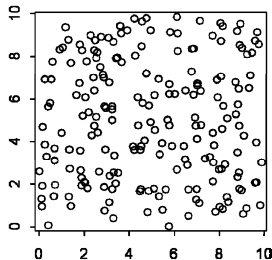
- ▶ **Lattice processes**: an area of spatial statistics concentrating on a feature class D with a countable (usually finite) number of member. That is, the **fixed** domain D is partitioned into a **countable** number of **lattices/areal units**
- ▶ The attributes are the random variables of interests



Types of Spatial Data

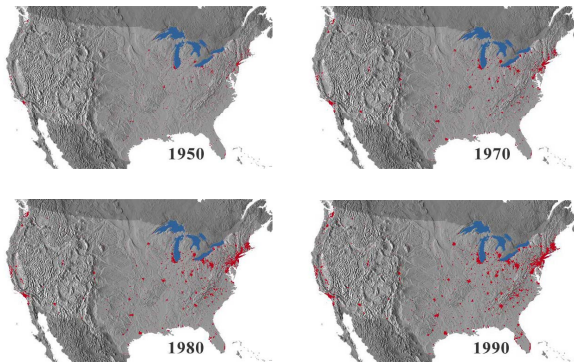
Point Patterns

- ▶ **Point processes:** an area of spatial statistics concentrating on a point feature class D with **random members**
- ▶ *The locations are the random variables of interests*



Types of Spatial Data

Spatio-temporal data



Urban areas are represented in red

Random Processes

- ▶ Processes that are subject to uncertainty; also called stochastic processes
- ▶ Governed by the laws of probability
 - ▶ Random variables (vectors)
 - ▶ (Joint) probability distributions
- ▶ Examples
 - ▶ Time series
 - ▶ Spatial random processes (random fields)
 - ▶ White noise: random variables are independent and identically distributed (iid)

Spatial Random Processes: Statistical model that specifies a joint probability distribution for the spatially indexed random variables

- ▶ Spatial random processes are represented by the **joint distribution** of a set of **(non-independent)** random variables Y indexed by location vector $s = (s_1, s_2)^T$ on study region D ; that is,

$$\{Y(s) : s \in D\}$$

- ▶ **Geostatistical data:** $D \subset \mathbb{R}^d$; D fixed
- ▶ **Lattice data:** $D = \{s_1, \dots, s_n\}$; D fixed
- ▶ **Point pattern:** $D = \{s_1, \dots, s_n\}$; n random, $s_1, \dots, s_n$ random

Specification of Spatial Random Processes

- ▶ Theoretically, to fully specify a spatial random process, we need to model the joint distributions of **every possible configurations of locations**

$f(Y(s_1), \dots, Y(s_m))$ for all $m = 1, 2, \dots$, and $s_1, \dots, s_m \in D$

- ▶ Real situation
 - ▶ In practice, we only have one realization from the joint distribution observed, which does not give much information about that distribution
 - ▶ We have to make some **“reasonable” assumptions** about the nature of the spatial phenomenon so that the model can be inferred using the observed data
 - ▶ As a consequence, we obtain a general mathematical form of the probability distribution (model) with some parameters whose values are left unspecified [Statistics: Data → Model]
- ▶ *Goal: to do inference (estimation, prediction, testing)*

Example: Accounting for dependence

- ▶ Let $Y_1, Y_2, \dots, Y_n$ be an iid sample from a normal distribution with unknown mean μ and known variance σ^2
 - ▶ The variance of $\bar{Y}$ is:
 - ▶ A 95% two-sided confidence interval for μ is:

- ▶ Now assume that for $i, j = 1, 2, \dots, n$

$$\text{cov}(Y_i, Y_j) = \sigma^2 \rho^{|i-j|}$$

- ▶ The variance of $\bar{Y}$ is

- ▶ Two 95% two-sided confidence intervals for μ are respectively:
(with $n = 10$, $\rho = 0.26$)

$$(\bar{Y} \pm (2.485)\sigma/\sqrt{10})$$

$$(\bar{Y} \pm (1.96)\sigma/\sqrt{10})$$

- ▶ In general, ignoring spatial dependence will lead to underestimation of the variance of estimators.

Summary

- ▶ Introduction to Spatial Statistics
- ▶ Types of Spatial Data
- ▶ Spatial Random Processes

Homework 0:

- ▶ Complete the last page of the syllabus and turn it in via Gradescope
- ▶ Getting started with R (and $\text{\LaTeX}$)

Preview:

- ▶ Structure of spatial random processes
- ▶ Variogram